

Shawn Yat Sin

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[linkedin.com/in/shawnyatsin](https://www.linkedin.com/in/shawnyatsin) | github.com/ShawnYS-codemtl | <https://shawnyatsin.github.io/full-stack-portfolio/>

SUMMARY

Full-stack developer experienced in Python, Java, and TypeScript, building scalable web applications with React, Next.js, and Node.js focused on intuitive user experiences and impactful solutions.

EDUCATION

McGill University *B. Sc. in Software Engineering* | GPA: 3.71/4.0 **2021 - 2025**

Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Design, Software Delivery, Software Validation, Operating Systems, Concurrent Programming

PROJECTS

[Sticker E-Commerce Platform](#) | *React, Next.js, TypeScript, Node.js, Tailwind CSS, Supabase (PostgreSQL), Stripe, Resend, Nodemailer, Vercel, Figma*

- Engineered and launched a production-grade full-stack e-commerce platform end-to-end, owning the full product lifecycle from UI/UX design in Figma to a responsive, SEO-optimized storefront built with Next.js and TypeScript.
- Architected secure backend workflows, modeling data in Supabase, implementing role-based admin access with Supabase Authentication and RLS, and integrating Stripe checkout with webhook-driven order processing.
- Accelerated operational efficiency by building admin dashboards for real-time order and inventory management, and automating customer email notifications across the full order lifecycle.

[Peer Mentorship Admin Platform](#) | *Python, Flask, React, TypeScript, SQLAlchemy, Tailwind CSS, Supabase (PostgreSQL), REST APIs, Vercel, Render*

- Reduced mentor-mentee assignment time from hours to seconds for 200+ participants by implementing a min-cost max-flow algorithm with multi-factor scoring across program, specialty, and identity dimensions.
- Deployed a full-stack application to production on Vercel and Render backed by Supabase PostgreSQL, with a two-step CSV import pipeline featuring auto-detected column mapping and structured validation feedback.
- Delivered an intuitive React/TypeScript admin dashboard empowering non-technical staff with real-time match score previews, manual override workflows, and match locking for full control over algorithmic assignments.

[Mars Rover Simulator](#) | *Python, Pygame, Rasterio, SQLite, PyInstaller, Jira*

- Collaborated in a 7-member Agile team, contributing to a Mars rover simulation platform using NASA HRSC MOLA terrain data, interactive maps, and path visualization.
- Implemented custom rover configurations and advanced pathfinding algorithms (A*, BFS, DFS) with configurable heuristics to support realistic mission planning.
- Designed and managed persistent project data using SQLite, tracking rover statistics and mission metrics including distance, energy usage, and duration.

[Smartly Goals](#) | *React, JavaScript, Node.js, Vercel Serverless Functions, HuggingFace Inference API*

- Built an interactive SMART goal-setting platform with an AI-powered review system, integrating the Llama 3 model to evaluate user-generated goals.
- Designed and refined real-time AI feedback logic by iterating on prompt structure and evaluation criteria to improve consistency and quality of goal assessments.
- Implemented PDF export functionality, enabling users to download polished, formatted goal summaries.

SKILLS

Programming languages: Python, JavaScript, TypeScript, Java, Swift, HTML, CSS, SQL

Frameworks/Libraries: Next.js, React, Node.js, Express.js, Tailwind CSS, Bootstrap, Flask, SQLAlchemy

Tools: Git, GitHub, Docker, Jira, Postman, Visual Studio Code, Xcode, PostgreSQL, SQLite, Unix, Supabase, Stripe, Figma, Vite, Claude Code, Vercel, Render

Languages: English, French

EXPERIENCE

Server (Part-time) - *Sansotei Ramen*

Aug 2024 - Present

- Operated effectively in a high-pressure, fast-paced environment, balancing customer communication, prioritization, and rapid context switching during peak service hours.
- Followed structured service processes with precision, demonstrating reliability, adaptability, and the ability to onboard quickly in a quality-critical workflow.

AI Language Contributor (Contract) - *Cheil Canada*

Dec 2023 - Jan 2024

- Evaluated and reviewed AI-generated French language content using detailed guidelines, ensuring linguistic accuracy and consistency across large sample sets.
- Identified subtle errors and edge cases across 200+ entries, demonstrating strong attention to detail and analytical judgment.

Sushi Chef (Part-time) - *Sushi Ya Matcha*

May 2022 - Oct 2022

- Executed time-critical, detail-oriented tasks with consistency and accuracy in a high-volume production environment.
- Collaborated closely with teammates to maintain quality standards under pressure, adapting quickly to changing demand.

Summer Camp Monitor - *Pointe-Claire Day Camps*

Jun 2021 - Aug 2021

- Led and supervised groups of 20+ children, prioritizing safety, conflict resolution, and clear communication.
- Took ownership of activity coordination, demonstrating leadership, accountability, and situational awareness in a team-based environment.

ACTIVITIES & LEADERSHIP

Webmaster, Nursing Undergrad Society (McGill University)

- Rebuilding the society's website from Wix to a custom solution to reduce operating costs, while maintaining a reliable online presence for its community.

Competitive Programming Participant (McGill University)

- Competed in university-level algorithmic contests including NP Compete, solving complex problems under strict time constraints using Python.

McGame Jam Participant (48-hour hackathon)

- Developed a game prototype in a 4-person team within 48 hours using JavaScript, HTML, and Phaser framework, focusing on rapid iteration and collaborative development under pressure.

VP Internal, Hong Kong Student Network (McGill University)

- Managed internal operations for a 30-member club, coordinating 3+ events per semester to drive student engagement and support executive initiatives.

Selected Participant, Ivey/IIBC Career Trek (Hong Kong)

- Chosen from a competitive cross-university applicant pool to engage with industry leaders across financial services, technology, and professional services, gaining exposure to international business practices and career pathways in the Asia-Pacific region.

Dragon Boat Team Member (McGill University)

- Trained and competed across two seasons, contributing to on-water synchronization and maintaining consistent commitment to team performance and conditioning.

Brown Belt, Judo

- Achieved through over 15 years of dedicated training, demonstrating discipline, perseverance, and continuous skill development.